

2-Mark Questions & Answers — MPCA Unit 2: Processor Organization & Architecture

Q1. Define pipelining.

Pipelining is an instruction execution technique where multiple instructions are overlapped in execution across independent hardware stages (Fetch, Decode, Execute, Memory, Write-back), increasing instruction throughput without increasing individual execution time.

Q2. What is a pipeline hazard?

A **Pipeline Hazard** is any condition or situation (such as structural resource contention, data dependencies, or branch instructions) that prevents the next instruction in the pipeline stream from executing in its designated clock cycle.

Q3. List Flynn's classification of computer architecture.

1. **SISD**: Single Instruction Stream, Single Data Stream.
 2. **SIMD**: Single Instruction Stream, Multiple Data Streams.
 3. **MISD**: Multiple Instruction Streams, Single Data Stream.
 4. **MIMD**: Multiple Instruction Streams, Multiple Data Streams.
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Q4. What is microprogrammed control unit?

A **Microprogrammed Control Unit** is a control unit design where control signals are generated by reading microinstructions (stored in a Control Memory ROM) sequentially, rather than using fixed hardwired logic gates.

Q5. Define instruction cycle.

An **Instruction Cycle** (or Fetch-Decode-Execute cycle) is the sequence of operations performed by the CPU to execute a single machine instruction. It consists of the **Fetch phase**, **Decode phase**, **Effective Address calculation phase**, **Execute phase**, and **Interrupt service check phase**.
